

A Report  
On the work done during 3<sup>rd</sup> semester subject  
Database Management Systems  
of  
B.Tech. Computer Engineering

## **INVENTORY MANAGEMENT SYSTEM**

|                                                                                                                                |                                                                                                                               |
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## Brief information about the system:

An inventory management system is the combination of technology (hardware and software) and processes and procedures that oversee the monitoring and maintenance of stocked products, whether those products are company assets, raw materials and supplies, or finished products ready to be sent to vendors or end consumers.

This system can widely be used by normal shops, departmental stores or MNCs for keeping a proper track of the stock. It also consists of information like manager details, customer details etc.

With the help of this system we can fix a minimum quantity of any inventory below which we need to place an order for that inventory.

This will help us in good sales results and never the out of stock stage for any inventory.

The project maintains three levels of users:

- Billing Counter Level
- Manager Level
- Owner Level

Main facilities available in this project are:

- We can forecast the sales by analyzing the previous sales statistics.
- We can get an idea that when we need to order new inventory.
- We can reduce the chances of any kind of frauds done by the staff members in the inventory.
- Customer details can be added.
- Invoice generation.
- We can keep a track of transactions received through different payment methods.

# Features:

## **INVENTORY CONTROL**

Inventory Control is organizing and monitoring of counted items on a retail sales floor, in storage, or in a warehouse. Good inventory control practices and systems make certain correct stock counts, proper product display and storage, accurate tagging, and accurate restock orders.

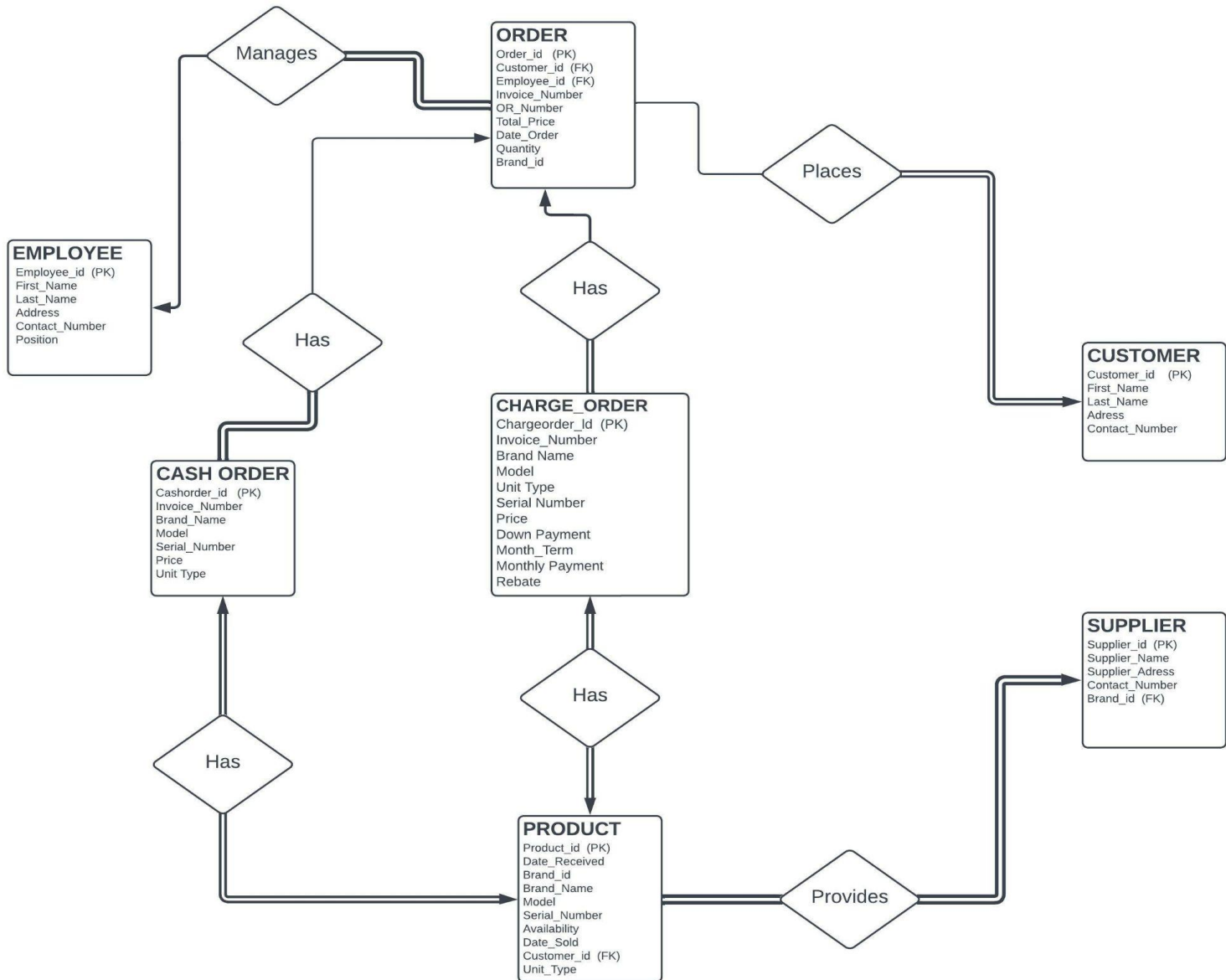
## **SALES REPORT**

Review Your Top-Line Revenue Performance. Eliminate order errors and duplicate facts entry with automation and 3rd party app integrations.

## **STOCK MONITORING**

Stock Monitoring keeps you from losing money with the price or stock changes. If the price on the supplier's site increases, it will raise it in your store accordingly, and if the price drops, it will also drop it in your store.

# ER Diagram:



[click to view](#)

## Schema:

**CUSTOMER**(Customer\_id (PK), First\_Name, Last\_Name, Address, Contact\_Number);

**ORDER**(Order\_id (PK), Customer\_id (FK), Employee\_id (FK), Invoice\_Number, OR\_Number, Total\_Price, Date\_Order, Quantity, Brand\_id);

**EMPLOYEE**(Employee\_id (PK), First\_Name, Last\_Name, Address, Contact\_Number, Position);

**CHARGE\_ORDER**(Chargeorder\_Id (PK), Invoice\_Number, Brand Name, Model, Unit Type, Serial Number, Price, Down Payment, Month\_Term, Monthly Payment, Rebate);

**CASH\_ORDER**(Cashorder\_id (PK), Invoice\_Number, Brand\_Name, Model, Serial\_Number, Price, Unit Type);

**SUPPLIER**(Supplier\_id (PK) , Supplier\_Name , Supplier\_Adress, Contact\_Number , Brand\_id (FK));

**PRODUCT**(Product\_id (PK), Date\_Received, Brand\_id, Brand\_Name, Model, Serial\_Number, Availability, Date\_Sold, Customer\_id (FK), Unit\_Type);

# CRUD:

## CREATE:

```
create table customer(  
    Customer_Id number(11),  
    First_Name varchar2(50),  
    Last_Name varchar2(50),  
    Address varchar2(50),  
    Contact_Number number(11)  
);
```

```
create table order(  
    Order_Id number(11),  
    Customer_Id number(11),  
    Employee_Id number(11),  
    Date_Order Date(30),  
    Invoice_Number number(11),  
    OR_Number number(11),  
    Total_Price number(50)  
);
```

```
create table employee(
```

```
Employee_Id number(11),  
First_Name Varchar2(50),  
Last_Name Varchar2(50),  
Address Varchar2(50),  
Contact_Number number(11),  
Position Varchar2(50)  
);
```

```
create table cash_order(  
    Cashorder_Id Number(11),  
    Order_Id number(11),  
    Brand_Name Varchar2(50),  
    Model Varchar2(50),  
    Serial_Number number(17),  
    Price Number(10),  
    Invoice_Number Number(11),  
);
```

```
create table charge_order(  
    Chargeorder_Id number(11),  
    Invoice_Number number(11),  
    Brand_Name varchar2(50),  
    Model varchar2(50),  
    Unit Type Varchar2(50),  
    Serial_Number Number(50),  
    Price Number(50),  
    Down_Payment Number(50),
```



```
Month_Term Number(50),  
Monthly_Payment Number(50),  
Rebate Number(50)  
);
```

```
create table supplier(  
    Supplier_Id number(11),  
    Supplier_Name Varchar2(50),  
    Supplier_Address Varchar2(50),  
    Contact_Number number(11),  
    Brand_Id number(11),  
);
```

```
create table products(  
    Product_Id number(11),  
    Date_Received Date(30),  
    Brand_Id number(11),  
    Brand_Name Varchar2(50),  
    Model Varchar2(50),  
    Serial_Number Number(50),  
    Availability Number(50),  
    Date_Sold Date(30),  
    Customer_Id Number(11),  
    Unit Type Varchar2(50)  
);
```

## **READ:**

### **customer order history**

```
select OR_Number, Total_Price, Date_Order, Quantity, brand_name, Model from  
order natural join product  
where customer_id=001;
```

### **invoice**

```
select * from charge_order , cash_order  
where invoice number=(select invoice number from order where order_id=001);
```

### **supplier information**

```
select supplier_name , Supplier_Adress, Contact_Number , brand_name , model  
from supplier as s, product as p  
where s.brand_id= p brand_id;
```

### **employee information**

```
select * from employee;
```

### **product sold data**

```
select brand_name, model , brand_id, quantity from product as p, order as o  
where p.brand_id = o.brand_id order by quantity desc;
```

### **availability**

select availability from product where product\_id=001;

### **customer's Monthly Payment**

select sum(monthly\_payment)from order as O , charge\_order as c  
where O.invoice number=C.invoice number and O.customer\_id==001;

## **UPDATE:**

### **update customer info**

update customer set address= xyz where customer\_id=001;  
update customer set contact\_number= xyz where customer\_id=001;

### **update employee info**

update employee set address= xyz where employee\_id=001;  
update employee set contact\_number= 9999999999 where employee\_id=001;  
update employee set position= abc where employee\_id=001;

### **update product info**

update product set availability=000 where product\_id=001;

## **DELETE:**

### **employee leaves the organization**

delete from employee where employee\_id=001;

### **supplier closed supplies**

delete from supplier where supplier\_id=001;

### **product is no longer available**

delete from product where product\_id=001;

# References:

1. **Lucidchart.com**(online ER diagram creation tool)
2. **stackoverflow.com**